

# AI for Marketing Decisions

Bridging Machine Learning with Strategic BI Command Architecture

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FINAL EXAM PROJECT

SPECIALTY: AI IN BUSINESS

ASTANA IT UNIVERSITY



# Executive Summary & Project Objectives

Transitioning from intuitive marketing to an algorithmic, single-source-of-truth **marketing command center** in Power BI — unifying five pillars of high-performance marketing.



## Behavioral Profiling

RFM-based customer classification



## Predictive Risk Mitigation

Churn probability via ensemble models



## Prescriptive Personalization

Collaborative filtering for cross-selling



## Sentiment Intelligence

NLP-driven social media polarity scoring

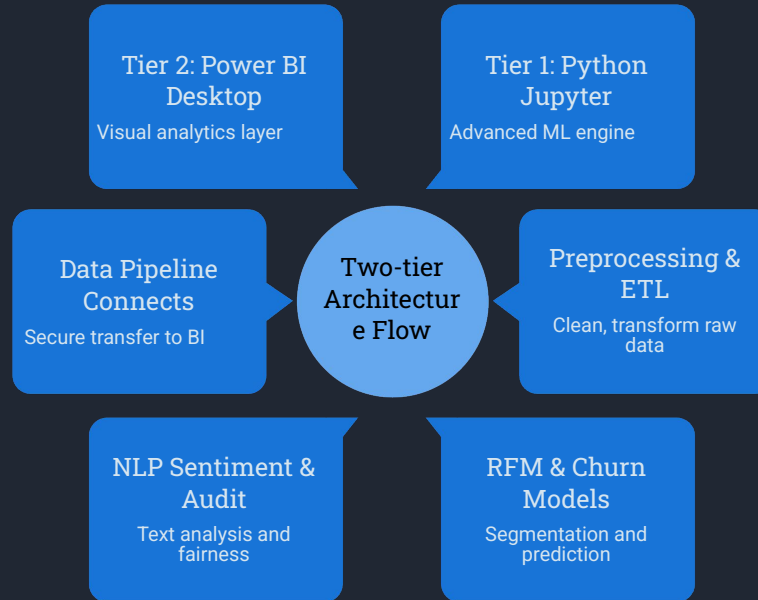


## Ethical ML Compliance

Fairness audits and GDPR adherence

# Project Architecture & Relational Schema

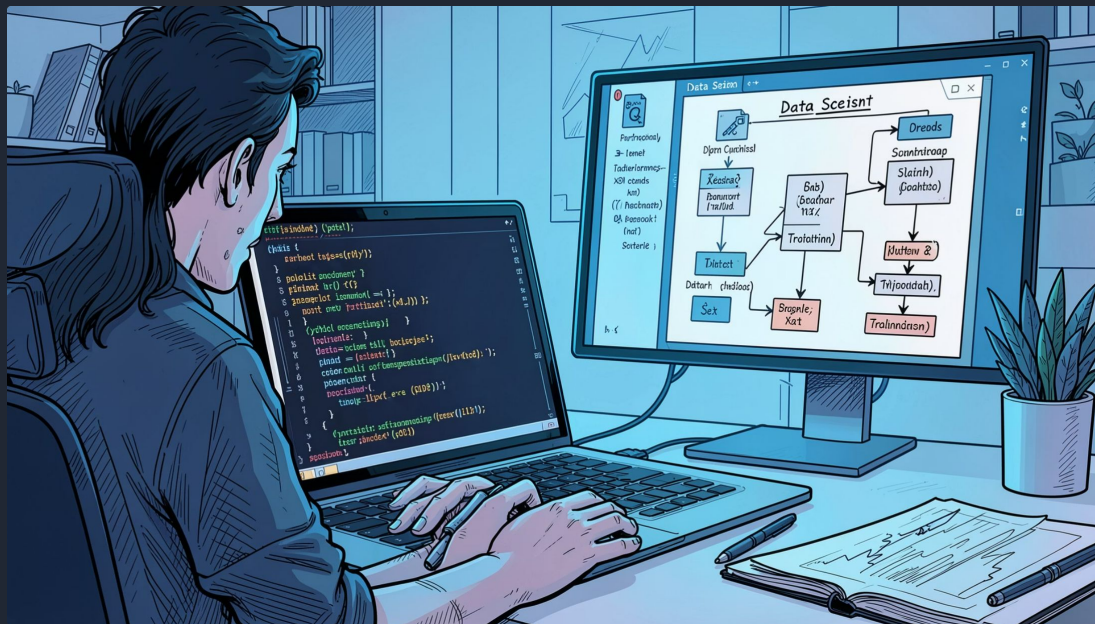
A two-tier architecture connects a Python ML Engine to a Power BI Visual Analytics Layer, with five relational tables forming the data model.



The pipeline flows from raw data processing in Python through to interactive executive dashboards in Power BI.



# Advanced Data & Feature Engineering Pipeline



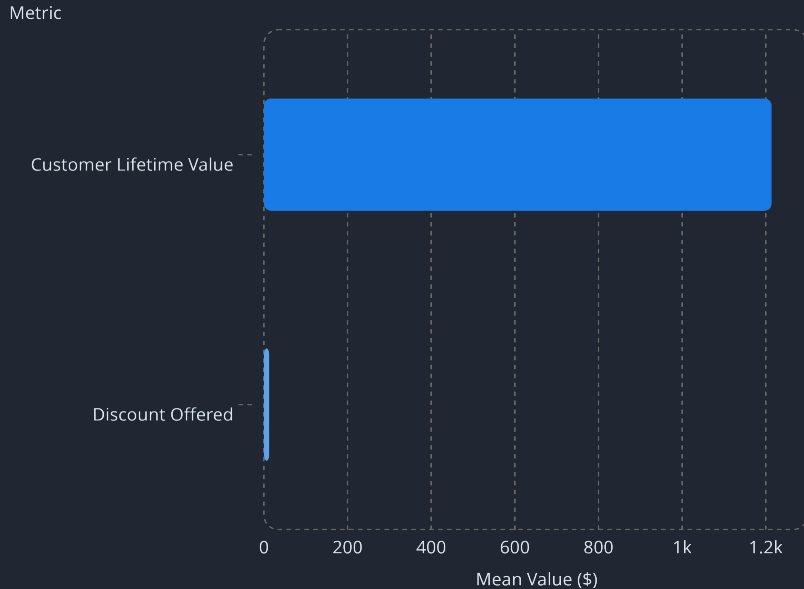
## Preprocessing Protocol

- **Median imputation** for missing values — preserving distribution integrity
- **Duplicate removal** across customer transaction logs
- **Data type casting** for modeling compatibility

## Engineered Financial Indicators

- **Average Order Value (AOV):** Revenue per transaction — measures purchase intensity
- **Customer Lifetime Value (CLV):** Cumulative revenue per account — identifies long-term

# Exploratory Data Analysis & Key Distributions



## CLV Distribution – Right-Skewed

Mean CLV of **\$1,214.4** with a right-skewed profile confirms that a small cohort of repeat buyers drives the majority of corporate revenue – validating VIP retention investment.

## Discount Distribution – Multi-Modal

Mean discount of **\$13.4** with a multi-modal pattern confirms distinct promotional campaign tiers operating simultaneously – enabling segment-specific offer optimization.



# Unsupervised Learning: RFM Behavioral Segmentation

RFM vectors replace rigid demographic buckets, classifying customers into three operationally distinct tiers for targeted intervention.

1

## High-Value Loyalists

Recent, frequent, high-spend — retain via **exclusive VIP programs** and early access

2

## At-Risk Core Spenders

Declining recency with strong history — trigger **automated win-back offers**

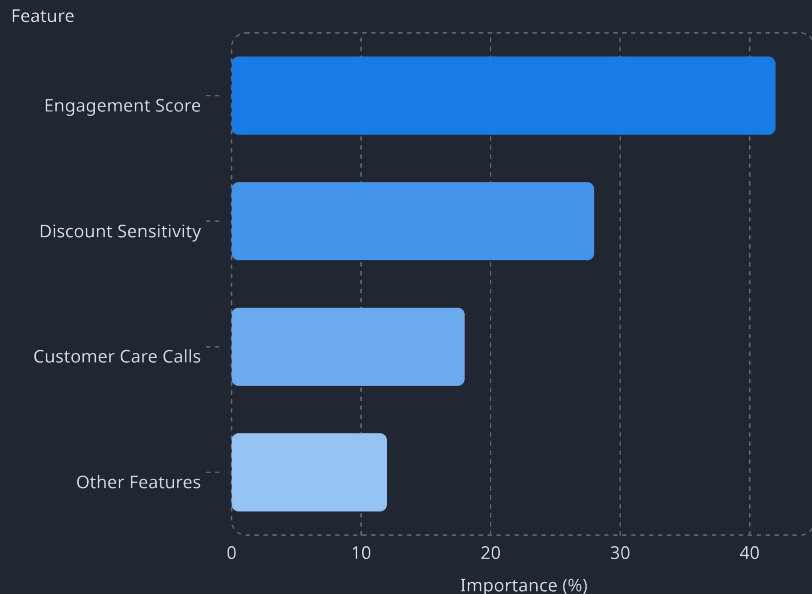
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## Bargain Hunters

High discount sensitivity, low frequency — deploy **margin-optimized flash sales**



# Supervised Learning: Predictive Churn Risk Modeling



## Ensemble Classification Model

Random Forest and XGBoost ensemble outputs **HVC\_Probability** — a continuous churn risk score per customer account, evaluated via Confusion Matrix for precision and recall.

## Top Predictive Drivers

- **Engagement Score (42%):** Dominant predictor of retention intent
- **Discount Sensitivity (28%):** Price dependency as churn signal
- **Customer Care Calls (18%):** Service friction as early-warning indicator

# Prescriptive Personalization & Sentiment Intelligence

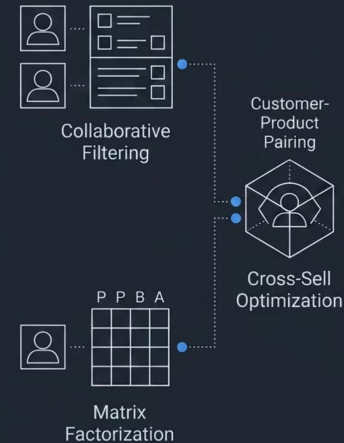
## Product Recommendation Engine

Collaborative filtering via **matrix factorization** dynamically pairs each Customer ID with top-predicted product categories — maximizing cross-sell potential and average basket size without manual rules.

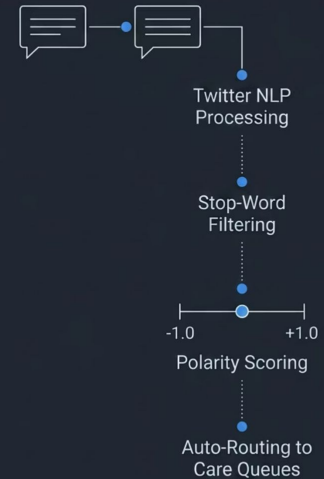
## NLP Sentiment Intelligence

Unstructured Twitter text is processed through stop-word filtering and tokenization, generating continuous **Polarity Scores (-1.0 to +1.0)**. Accounts exceeding negative thresholds are auto-routed to customer care queues for proactive intervention.

## Recommendation Engine



## Sentiment Intelligence







## DIGITAL COMPLIANCES

# Algorithmic Governance & Responsible AI Compliance

### Algorithmic Fairness Audit

The **Disparate Impact Ratio (DIR)** is tracked across protected demographics within the `fairness_summary` layer. Compliance threshold: **DIR between 0.80 and 1.25** — ensuring no segment experiences systematically biased model outcomes.

### GDPR Data Compliance

All **Personally Identifiable Information (PII)** undergoes complete anonymization and cryptographic hashing at the database source layer — prior to any ML processing — ensuring full regulatory adherence and auditability.

# Strategic Recommendations & Implementation Roadmap

01

## Phase 1 — Pipeline Cleansing (Days 1–30)

Validate data quality; deploy PII hashing

02

## Phase 2 — Pilot Testing (Days 31–60)

Launch churn model and RFM segmentation on live data

03

## Phase 3 — Full Automation (Days 61–90)

Automate interventions, recommendations, and sentiment routing

## Actionable Takeaways

- Trigger automated risk interventions for high-churn accounts
- Inject personalized affinity mappings into marketing channels
- Optimize margins for price-inelastic loyalist segments
- Establish sentiment-driven support routing workflows



**Project Portfolio:** Access full source code, preprocessed datasets, and complete technical report at:

<https://github.com/ilyas06089-collab/Ilyas>